

COMPUTER SCIENCE CURRICULA 2023: REPORT BY THE ACM/IEEE-CS/AAAI JOINT TASK FORCE

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Association for
Computing Machinery

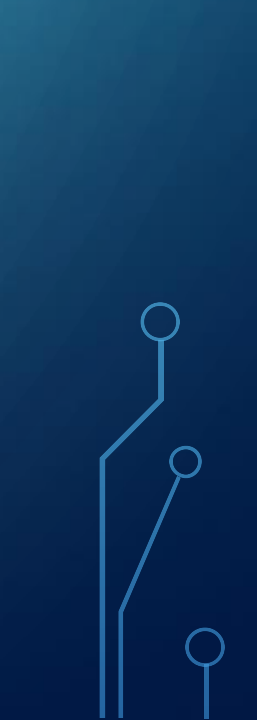


IEEE
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SOCIETY

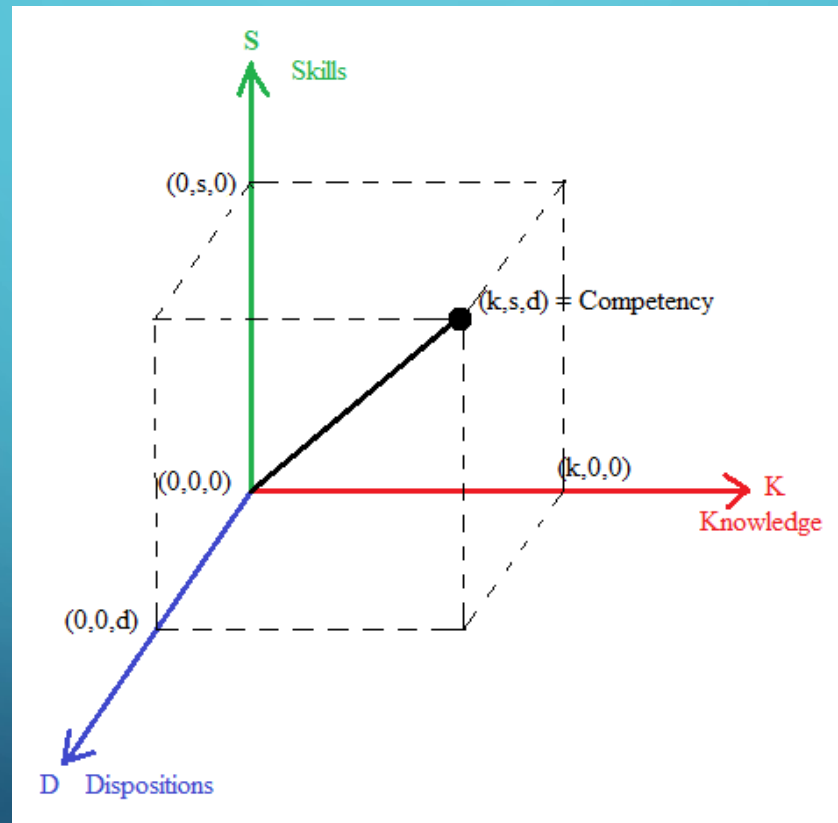




CONTENTS

- **Introduction**
 - **Highlights**
 - **Knowledge Model**
 - **Competency Model**
 - **Curricular Practices**
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COMPETENCY = KNOWLEDGE + SKILLS + DISPOSITIONS



COMPETENCY = KNOWLEDGE + SKILLS + DISPOSITIONS

- Knowledge Model = { Knowledge Areas }
- Knowledge Area = { Knowledge Units } + { **Professional Dispositions** }
- Knowledge Unit = { Topics } + { **Illustrative** Learning Outcomes }
- Topic $\in$ { **CS Core** } $\cup$ { **KA Core** } $\cup$ { Non-core }
- **Competency area** $\subseteq$ Knowledge model

COMPETENCY = KNOWLEDGE + SKILLS + DISPOSITIONS

- Explain
- Apply
 - Generative AI
- Evaluate
- Develop

COMPETENCY = KNOWLEDGE + SKILLS + DISPOSITIONS

- ▣ Malleable values, beliefs and attitudes that enable consistent behaviors desirable in the workplace
- ▣ Willingness and intent to apply knowledge + skills
 - Teachable → Learnable
 - Measurable → Observable
 - Summative → Formative
 - Discipline-specific
 - Task-specific

DISPOSITIONS [CC 2020]

- Adaptable
 - Collaborative
 - Meticulous
 - Persistent
 - Responsive
 - Self-Directed
- *Not exhaustive*

COMPETENCY MODEL

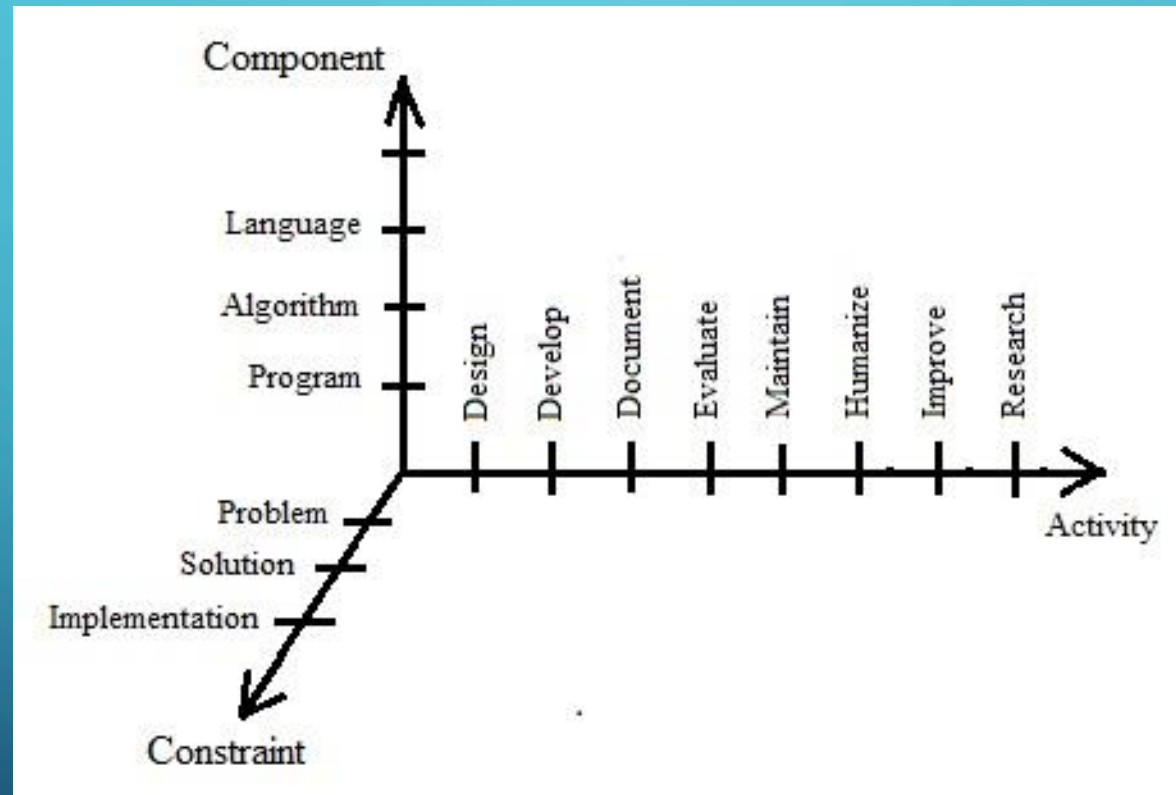
- Competency Model = { Competency specifications }
- Competency specification
 - **Task** + Competency statement + Knowledge + Skills + Dispositions
 - Dispositions: Identified for each knowledge area

COMPETENCY FRAMEWORK

- A framework for **task specification**
- A **format** for Competency specification
- An **algorithm** for curriculum design using the Competency Framework

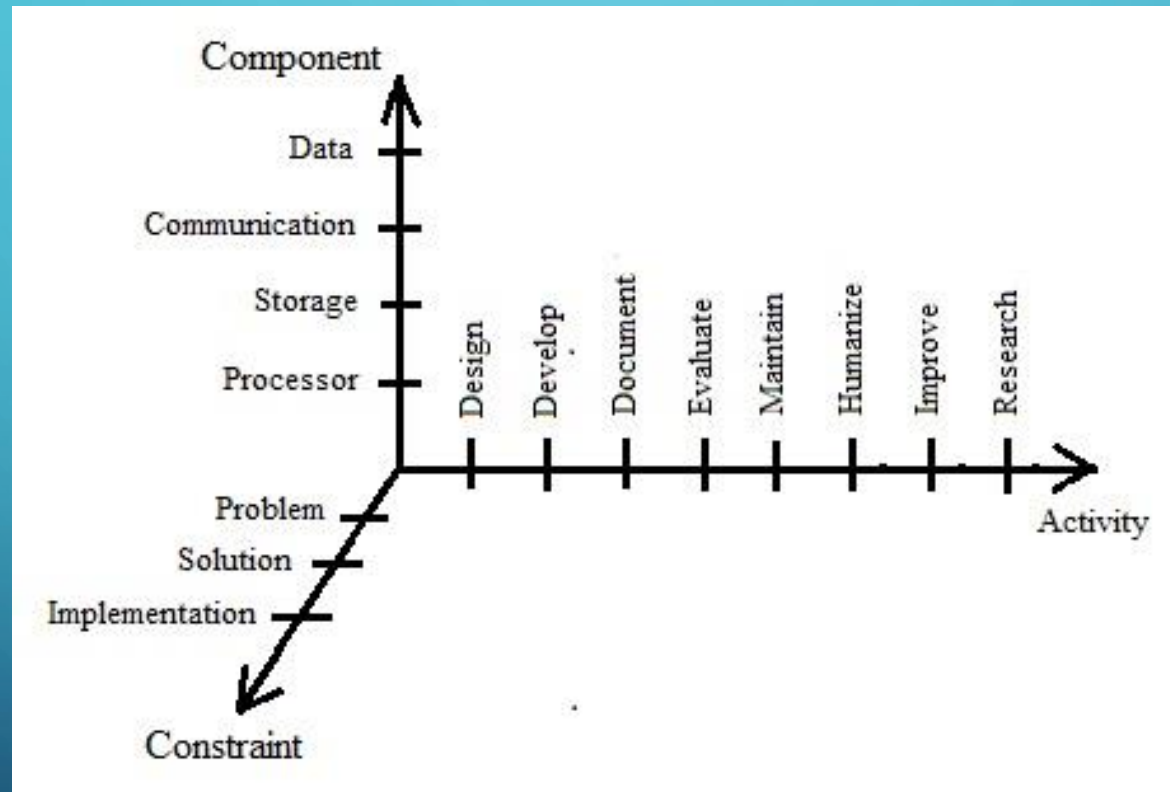
TASK SPECIFICATION

– SOFTWARE COMPETENCY AREA



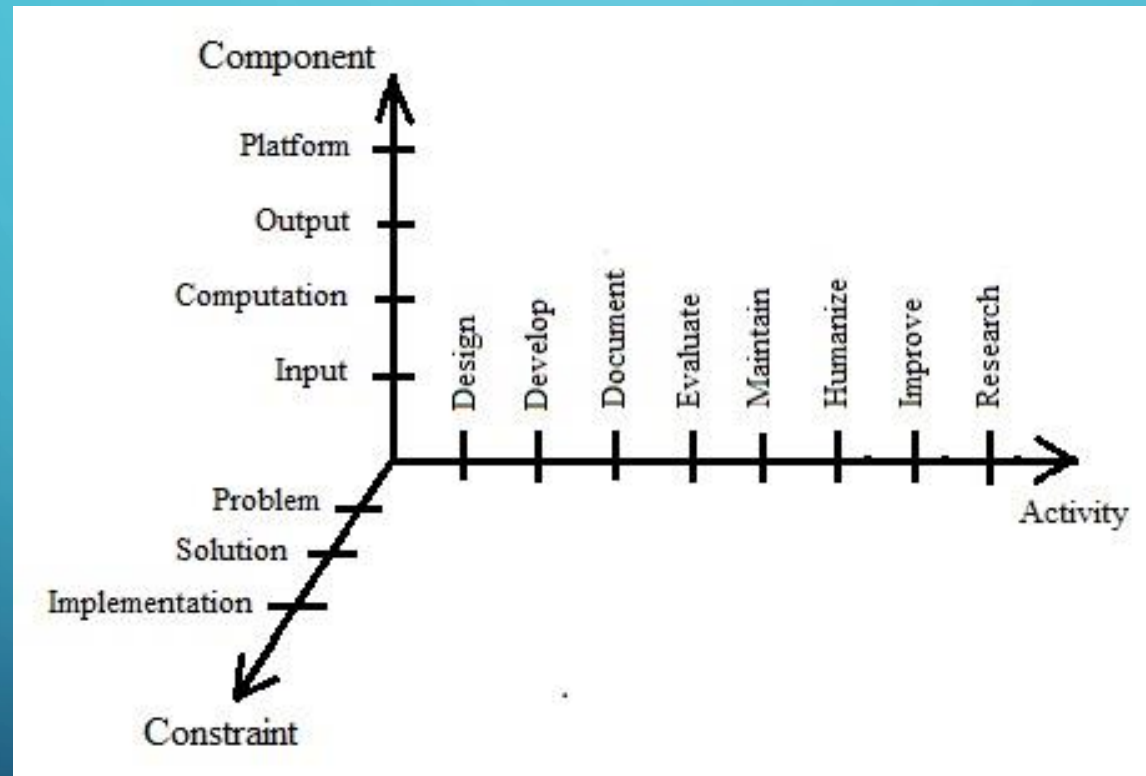
TASK SPECIFICATION

- SYSTEMS COMPETENCY AREA



TASK SPECIFICATION

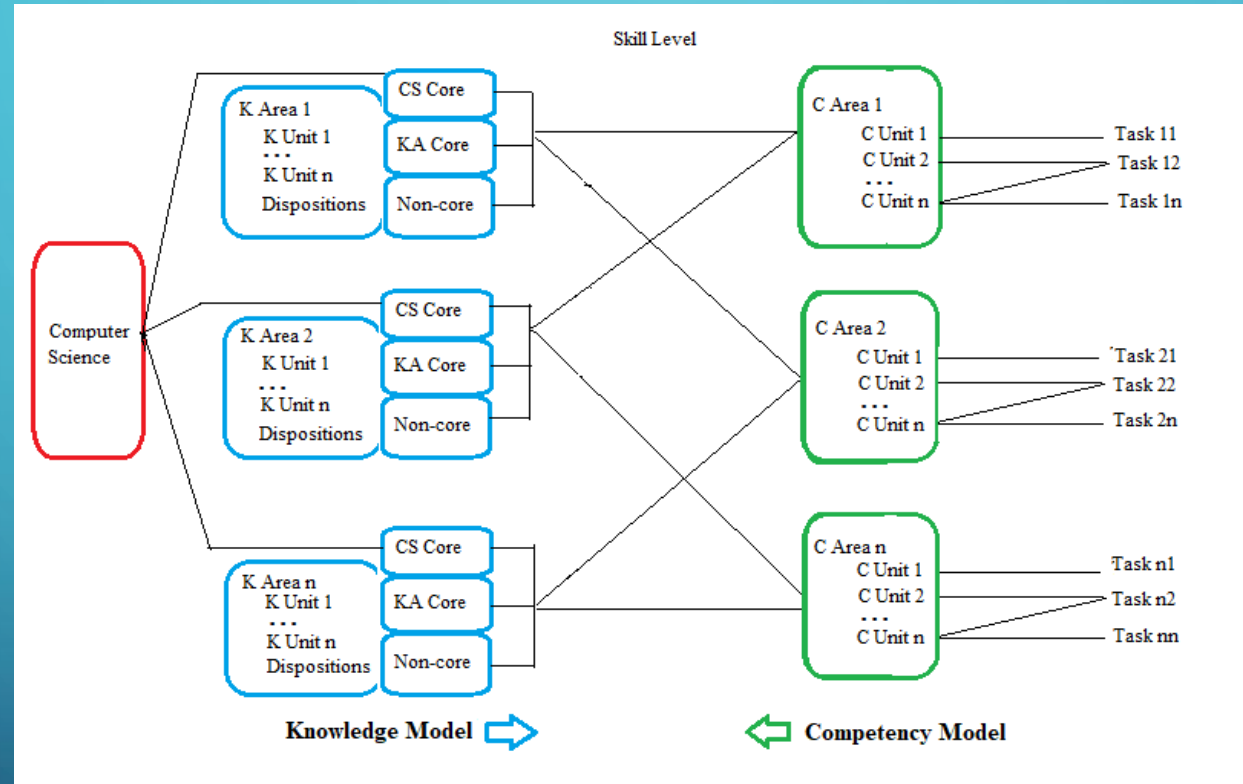
- APPLICATIONS COMPETENCY AREA



COMPETENCY SPECIFICATION

- **Task:** Non-technical, high level
 - Only independent variable
- **Competency statement:** Technical
- **Required knowledge:** KA / KU
- **Required skill level:** Explain / Apply / Develop / Evaluate
- **Professional Dispositions:** Adaptable / Collaborative / Meticulous / ...

INTEGRATED MODEL



CURRICULUM DESIGN FROM COMPETENCY FRAMEWORK

Atomic Task: Activity X Component X Constraint

→ Compound Task

→ { Competency specifications } = Competency Model

→ Knowledge Area-Knowledge Unit

→ Course Packaging

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- **Curricular Practices**

ARTICLES ON CURRICULAR PRACTICES

▣ Social aspects

▣ Accessibility

▣ Ethics

▣ Ethics in the Global Souths

▣ CS 4 Good

ARTICLES ON CURRICULAR PRACTICES

▶ Professional practices

- ▶ Liberal Arts Perspective

- ▶ Community College Perspective

- ▶ Geographic Regions – *ACM Inroads*:

 - ▶ Arab World / Africa / China / Philippines / Australia / Latin America

ARTICLES ON CURRICULAR PRACTICES

▶ Programmatic considerations

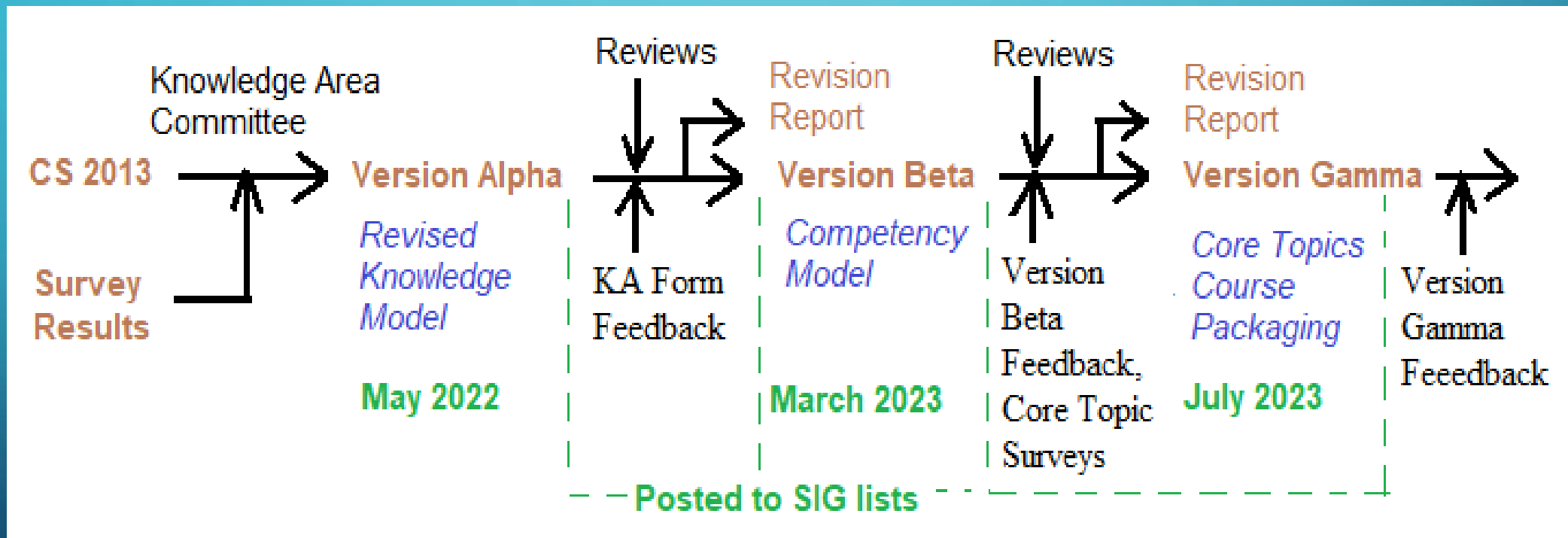
- ▶ CS+X

- ▶ Formal methods in CS education

- ▶ Large Language models in introductory programming

- ▶ Quantum computing education

TIMELINE



LOOKING AHEAD

- Copy-edited
- Updated as a web site
- Visualization tool
- Machine-readable format



ADDITIONAL INFORMATION

csed.acm.org

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